222724: metagranodiorite, Gregory Well

(Yilgarn Craton granites, Kalgoorlie Terrane, Eastern Goldfields Superterrane, Yilgarn Craton)

Location and sampling

Leonora (SH 51-1), Wilbah (3040)
MGA Zone 51, 236185E 6791942N

Sampled on 8 April 2018

This sample was collected from an outcrop northeast of the Mount Ida mining centre, about 5.9 southwest of Windy Bore, 5.5 km north of the Leonora – Mount Ida Road, and 5.4 km east of Gregory Well.

Geological context

The unit sampled is a non-deformed metagranodiorite of the Kalgoorlie Terrane, and is located to the east of the eastern limb of the Kurrajong Anticline in the Mount Ida greenstone belt (Stewart, 2004). Nearby dated rocks are within the eastern limb of the Kurrajong Anticline. The Forest Belle Gabbro, sampled about 9.4 km to the southwest, yielded an igneous crystallization age of 2731 ± 8 Ma (GSWA 222720, Wingate et al., 2022b). About 6.6 km to the southwest, quartzite interleaved with mafic–ultramafic rocks of the Walter Williams Formation of the Kalgoorlie Group (Swager et al., 1990; Hill et al., 2001), yielded a maximum depositional age of 2764 ± 8 Ma (GSWA 222712, Wingate et al., 2022a).

Petrographic description

The sample is a strongly weathered, medium- to coarse-grained metagranodiorite (Fig. 1), consisting of about 45–50% ex-plagioclase, 20% K-feldspar, 10–20% quartz, 12% ex-ferromagnesian minerals, 2–3% muscovite, and trace opaque minerals. Original plagioclase is anhedral to subhedral, up to 2.5 mm in size, and almost completely replaced by extremely fine-grained, pale brown minerals (possibly clinozoisite or clay), which are variably overprinted by very fine-grained carbonate minerals. A few areas of relict plagioclase exhibit poorly defined multiple twinning. K-feldspar is anhedral, fractured, up to 5 mm in size, slightly altered to clay and carbonate minerals, and exhibits microcline twinning. Pale brown, equant to tabular grain shapes, up to 2.5 mm in size, and minor patches of pale yellow epidote, are interpreted as probable ex-ferromagnesian grains. Quartz is anhedral, irregular, up to 7 mm long, and fractured but not strained. Subhedral probable muscovite laths up to 0.7 mm long may have replaced original biotite, and locally occur in aggregates. Opaque minerals are equant, anhedral to subhedral, up to 0.45 mm in size, and probably magnetite replaced by hematite and titanite. The sample lacks a foliation and it is not possible to determine the degree of metamorphism due to the almost complete obliteration of minerals other than quartz and K-feldspar. Brittle deformation has resulted in extensive fracturing.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown, anhedral to subhedral, and variably fractured and broken. The crystals are up to 200 µm long, and equant to elongate, with aspect ratios up to 4:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, and most crystals are dominated by high-uranium, metamict zones. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 23–24 July 2020, using SHRIMP-B. Twelve analyses of the M257 standard obtained during the session indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.18% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Thirteen analyses were obtained from 11 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). One analysis older than 2800 Ma is >5% discordant. The date obtained from this analysis (Group D; Table 1) is unreliable, and considered not to be geologically significant. The remaining 12 analyses can be divided into three groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 10 analyses of eight zircons (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2648 ± 8 Ma (MSWD = 0.79).

Group X comprises one analysis (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 2705 ± 14 Ma (1σ).

Figure 1. Hand specimen image for sample 222724: metagranodiorite, Gregory Well

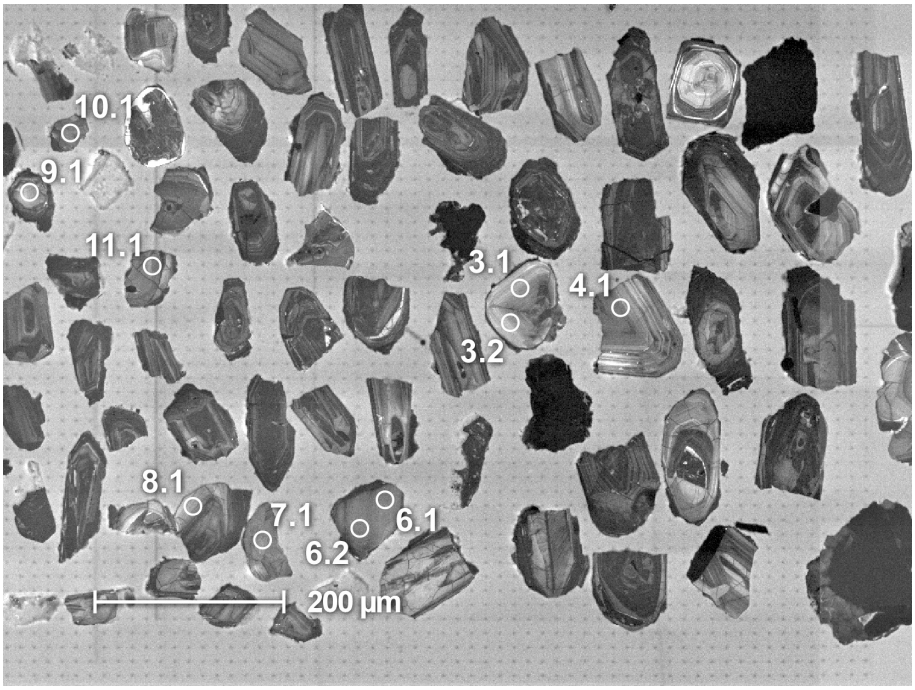

Figure 2. Cathodoluminescence image of representative zircons from sample 222724: metagranodiorite, Gregory Well. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 222724: metagranodiorite, Gregory Well

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	1	1.1	124	135	1.09	0.207	2.051	0.024	0.17925	0.00116	2.056	0.024	0.17740	0.00129	2555	25	2629	12	2.8
I	6	6.1	147	44	0.30	0.017	1.966	0.022	0.17817	0.00107	1.967	0.022	0.17802	0.00108	2650	24	2635	10	-0.6
I	4	4.1	120	252	2.10	0.978	2.557	0.050	0.18772	0.00133	2.582	0.051	0.17899	0.00214	2110	36	2644	20	20.2
I	5	5.1	53	83	1.57	0.646	2.368	0.038	0.18481	0.00178	2.384	0.038	0.17905	0.00233	2258	31	2644	22	14.6
I	9	9.1	105	109	1.03	-0.045	2.051	0.026	0.17919	0.00125	2.050	0.026	0.17959	0.00129	2561	27	2649	12	3.3
I	13	3.2	129	243	1.88	0.424	2.183	0.025	0.18365	0.00122	2.193	0.025	0.17987	0.00148	2422	23	2652	14	8.6
I	10	10.1	160	219	1.37	0.695	2.326	0.025	0.18649	0.00113	2.342	0.025	0.18030	0.00161	2292	21	2656	15	13.7
I	12	6.2	166	58	0.35	-0.028	2.025	0.021	0.18017	0.00099	2.024	0.021	0.18041	0.00101	2588	23	2657	9	2.6
I	3	3.1	83	74	0.90	0.171	2.002	0.027	0.18213	0.00143	2.006	0.028	0.18060	0.00156	2608	30	2658	14	1.9
I	7	7.1	121	82	0.68	0.246	2.257	0.027	0.18310	0.00128	2.262	0.027	0.18090	0.00145	2360	24	2661	13	11.3
X	8	8.1	82	67	0.81	0.059	1.982	0.028	0.18629	0.00150	1.983	0.028	0.18576	0.00154	2632	30	2705	14	2.7
P	2	2.1	115	101	0.88	0.452	2.050	0.025	0.17938	0.00120	2.059	0.025	0.17535	0.00148	2552	26	2609	14	2.2
D	11	11.1	189	185	0.98	0.275	1.984	0.020	0.20092	0.00100	1.990	0.020	0.19847	0.00113	2625	22	2814	9	6.7

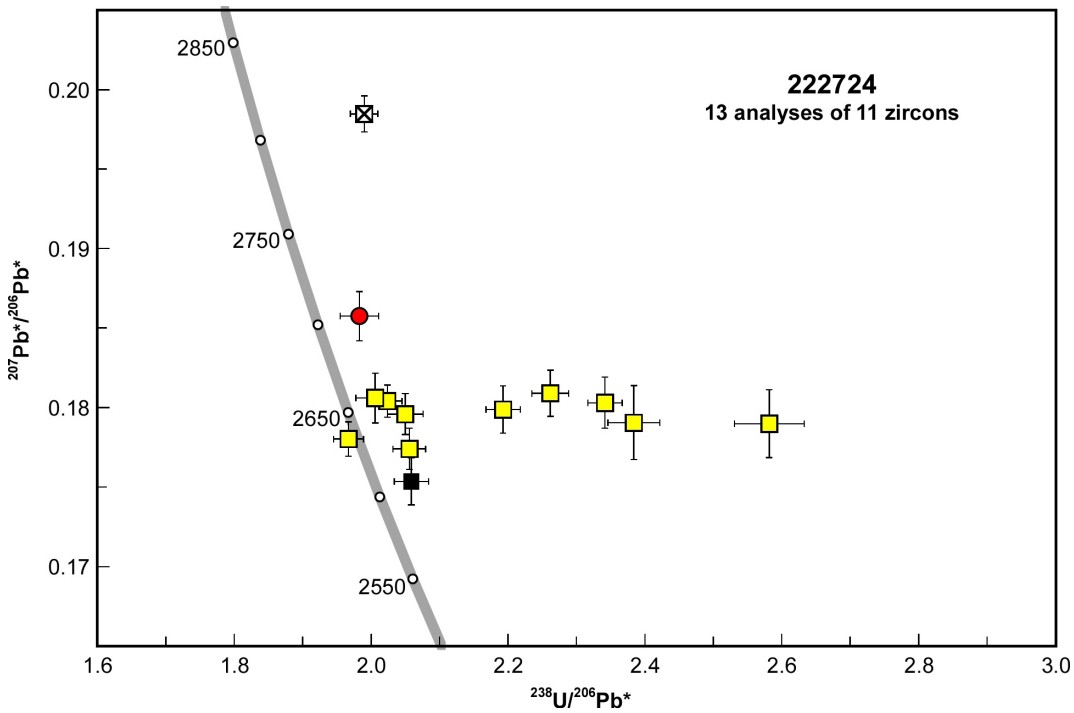

Figure 3. U–Pb analytical data for zircons from sample 222724: metagranodiorite, Gregory Well. Yellow squares indicate Group I (magmatic zircons); red circle indicates Group X (xenocrystic zircon); black square indicates Group P (radiogenic-Pb loss); crossed square indicates Group D (date >2800 Ma and discordance >5%)

Group P comprises one analysis (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 2609 ± 14 Ma (1σ).

The date of 2648 ± 8 Ma for the 10 analyses in Group I is interpreted as the igneous crystallization age. The date of 2705 ± 14 Ma (1σ) for the single analysis in Group X is interpreted as the age of an inherited zircon. The date of 2609 ± 14 Ma (1σ) for the single analysis in Group P is interpreted to reflect minor ancient loss of radiogenic Pb.

References

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Recommended reference for this publication

Wingate, MTD, Lu, Y, Fielding, IOH and Sapkota, J 2022, 222724: metagranodiorite, Gregory Well; Geochronology Record 1884: Geological Survey of Western Australia, 4p.

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